EE2004: Digital Signal Processing

Tutorial 11
(April 27, 2026)

1. Let $x[\cdot] = \{\overset{\downarrow}{3}, 2, 3\}$. Find the 3-point DFT of $x[n]$.

2. Let $x[\cdot] = \{\overset{\downarrow}{3}, 0, -2, 2\}$. Find the 4-point DFT of $x[n]$.

3. The 8-point DFT of a real-valued signal $x[n]$ is known over $0 \leq k \leq 4$ to be $X[k] = 1 + 2j\,\delta[k-2] + \delta[k-4]$.
   (a) Determine $X[k]$ for $5 \leq k \leq 7$.
   (b) Determine $x[n]$.

4. Consider a complex signal consisting of two closely spaced sinusoids, i.e., $x[n] = e^{j2\pi(0.30)n} + e^{j2\pi(0.33)n}$. For each of the cases below, plot the $|X[k]|$ as a function of $f_k = k/N$.
   (a) Compute and plot the 10-point DFT of $x[n]$ using 10 samples ($0 \leq n \leq 9$). From the plot, can both exponentials be identified? Explain.
   (b) Zero-pad the signal from (a) with 490 zeros and computed the 500-point DFT. Does this improve the picture of the DFT. Explain.
   (c) Compute and plot the 100-point DFT of $x[n]$ using 100 samples ($0 \leq n \leq 99$). From the plot, can both exponentials be identified? Explain.
   (d) Zero-pad the signal from (a) with 400 zeros and computed the 500-point DFT. Does this improve the picture of the DFT. Explain.

5. Repeat the previous problem when $x[n] = e^{j2\pi(0.300)n} + e^{j2\pi(0.315)n}$

6. Consider the length-4 unit amplitude pulse $x[n] = u[n] - u[n-4]$.
   (a) Compute and plot the 4-point DFT $X[k]$ of the signal $x[n]$.
   (b) Use the DFT interpolation formula to compute the DTFT $X(e^{j\omega})$ from the DFT $X[k]$. Sample this $X(e^{j\omega})$ over $0 \leq \omega \leq 2\pi$ using 10 equally spaced points to create $X_{10}[k]$. Compute and plot the IDFT of $X_{10}[k]$. Comment on your results.

7. Using only the properties of the DFT and the fact that $\delta[n] \longleftrightarrow 1$, determine the 4-point DFTs of the following sequences: (a) $x_a[\cdot] = \{\overset{\downarrow}{0}, 2, 0, 0\}$, (b) $x_b[\cdot] = \{\overset{\downarrow}{0}, 0, 0, 4\}$, (c) $x_c[\cdot] = \{\overset{\downarrow}{0}, 0, j, 0\}$, (d) $x_d[\cdot] = \{\overset{\downarrow}{0}, 4, -j, 2\}$.

8. Find $x_1[n] \circledast_4 x_2[n]$ where $x_1[\cdot] = \{\overset{\downarrow}{1}, -2, 3, -4\}$ and $x_2[\cdot] = \{\overset{\downarrow}{1}, -1, -2, 2\}$.

9. For the previous problem what is the result of linear convolution? For what value of N do the results of circular convolution and linear convolution match over $0 \leq n \leq N-1$.

10. Let $x_c(t) = \cos(2\pi F_0 t)$ where $F_0 = 1\,\text{kHz}$. This signal is sampled at a frequency $F_s = 8\,\text{kHz}$. Let $x[n]$ denote the samples of $x_c(t)$ taken over two cycles.
    (a) Compute the 16-point DFT and plot the magnitude (stem plot). In these and other plots, the x-axis must be in Hz over the range 0 to 4000 Hz.
    (b) Repeat the above part for the following lengths (after zero-padding): 32 and 64. Overlay these results (using different colours) on the stem plot for the 16-point DFT.
    (c) Now zero-pad $x[n]$ such that the total length is 8192. Overlay the normal magnitude plot (don't use stem plot for this part) on the previous plots. Are you able to see that you can view the DFT as the samples of the DTFT?

11. Let $x[n]$ be the same sequence that was defined in the previous problem.
    (a) Suppose we want to consider, as before, just two cycles of $x[n]$. Over what range of values n should $x[n]$ be defined if you want the DFT to be real-valued? How should these samples be arranged to form the vector `x` for the MATLAB command `X = fft(x)`. Verify that the DFT coefficients are indeed real-valued (or, at best, the imaginary parts are negligibly small, caused by round-off errors).
    (b) Repeat the previous part for $N = 32$ and $N = 64$. That is, $x[n]$ continues to be made of two cycles, but zero-padded appropriately to get the 32-point and 64-point DFTs. We require the DFTs to be real-valued for each of these lengths. How should the samples of $x[n]$ be arranged in each case? ◇